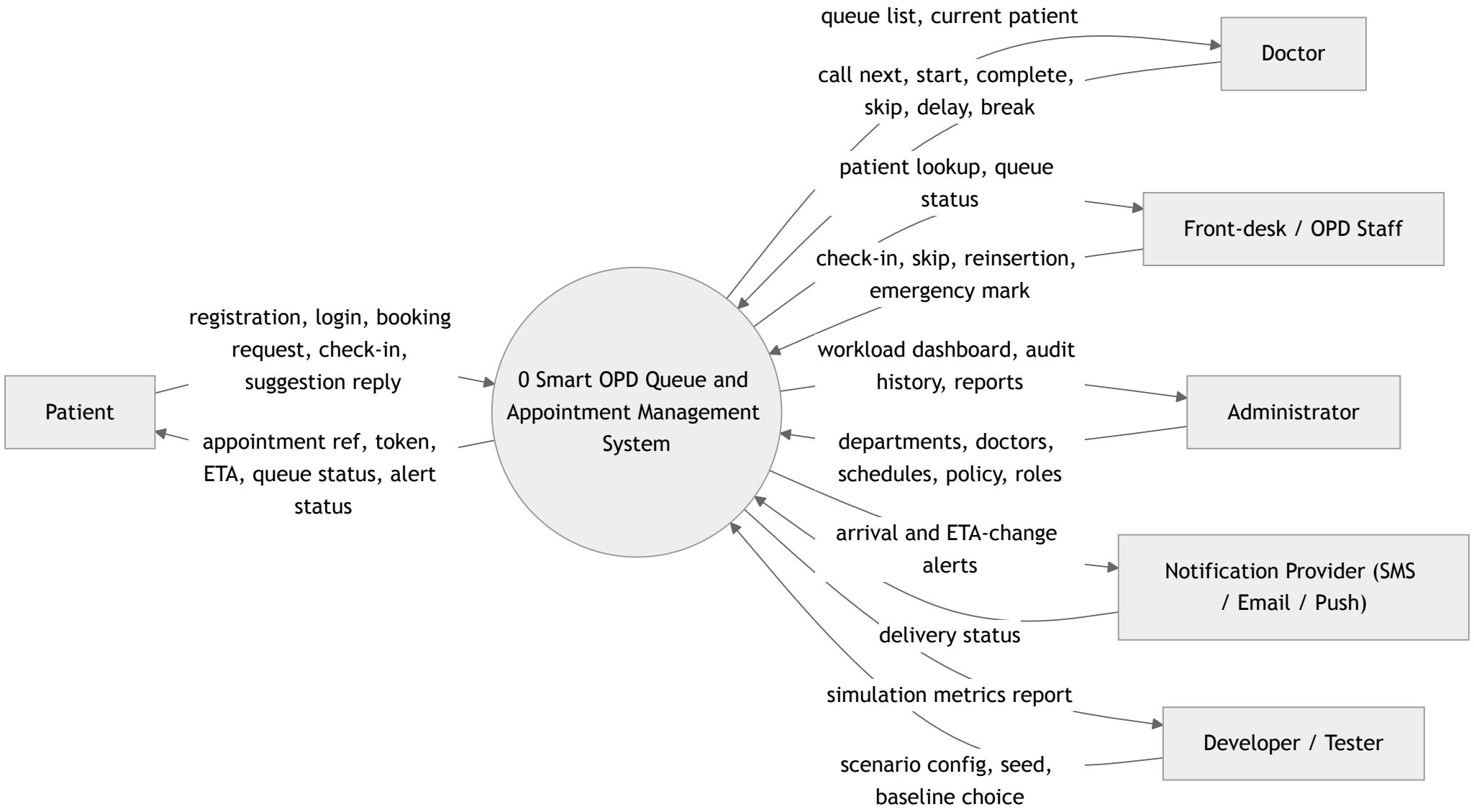
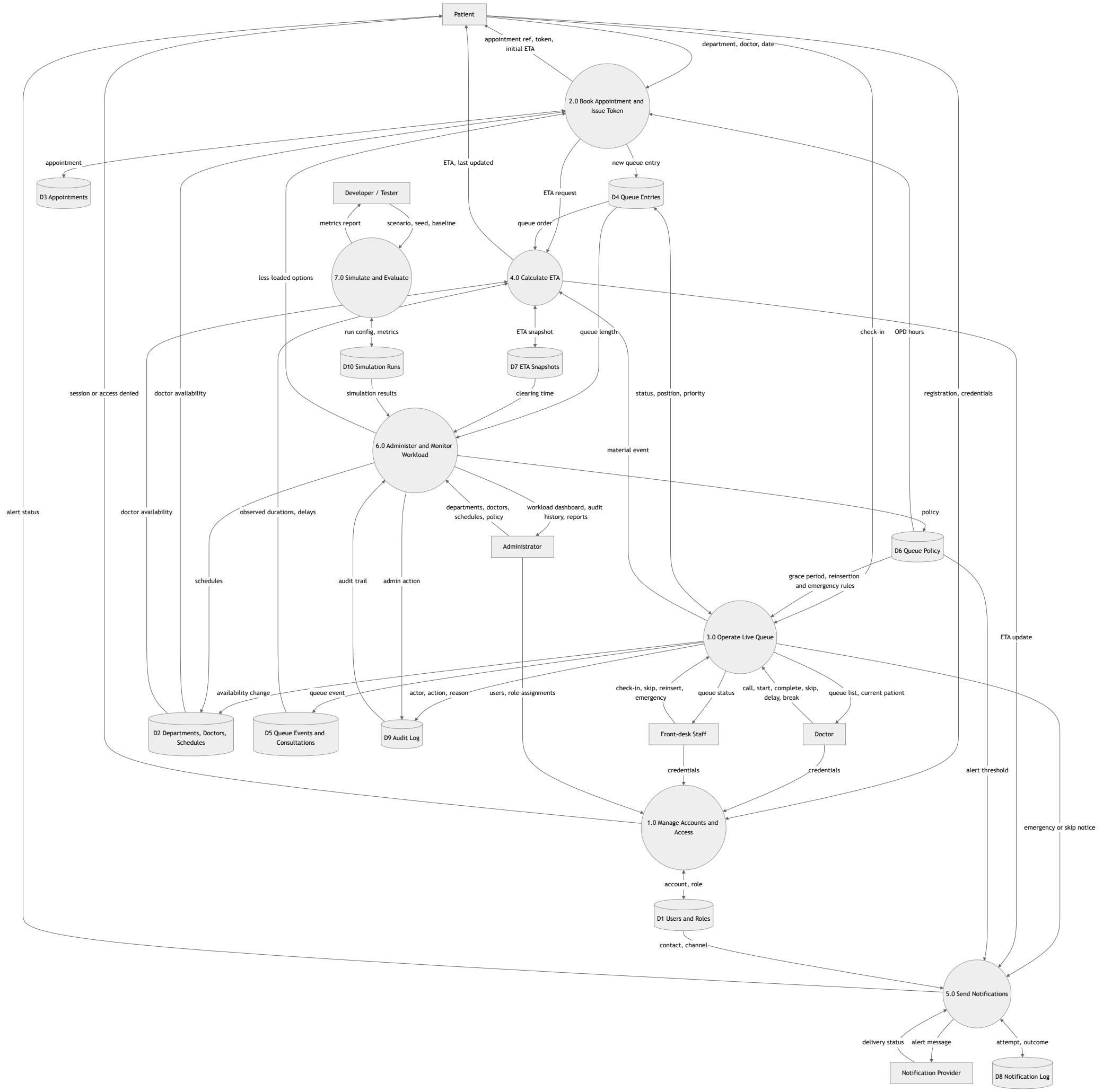
Smart OPD Queue and Appointment Management System

DFD Level 0 (Context Diagram)

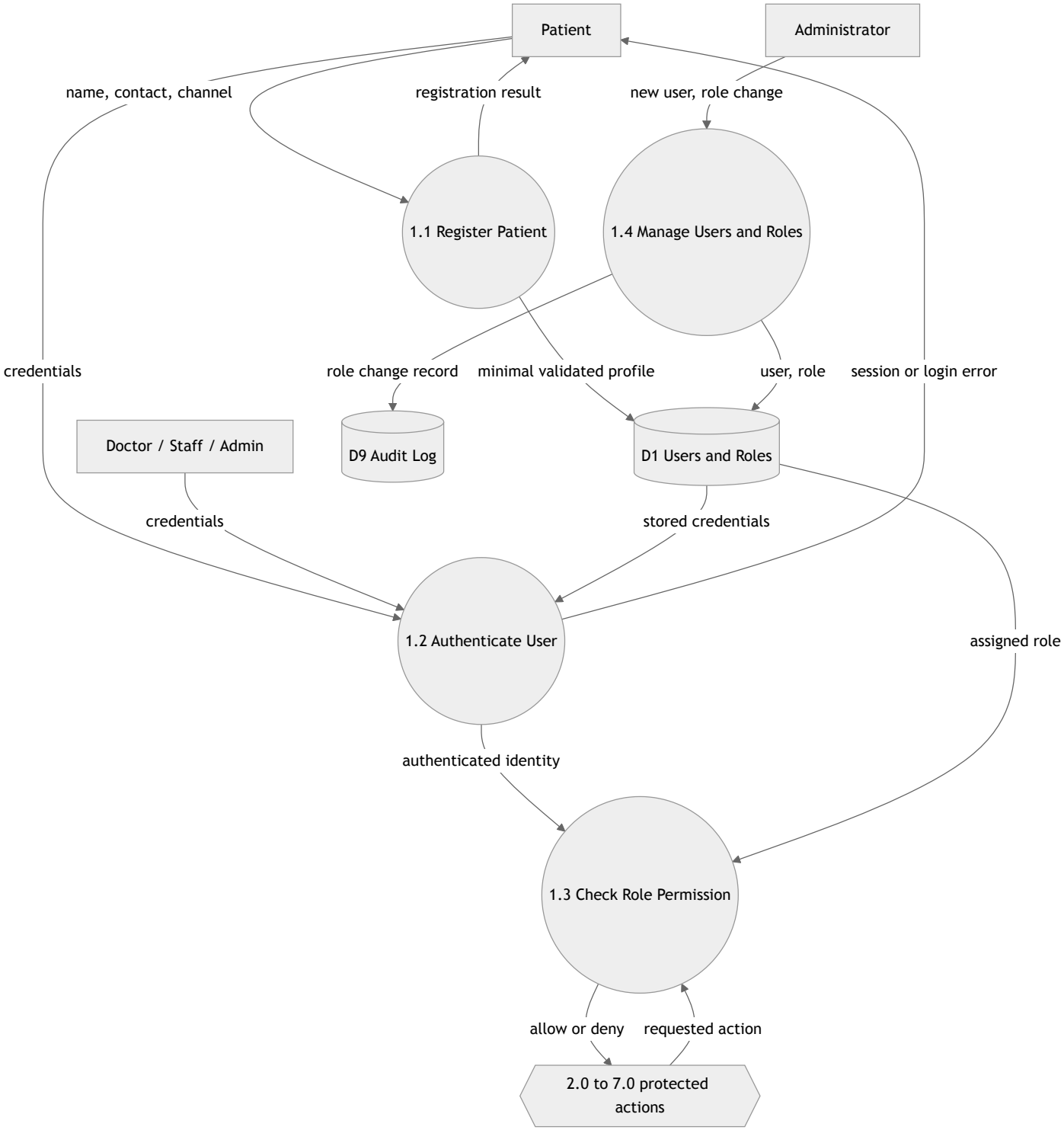


DFD Level 1

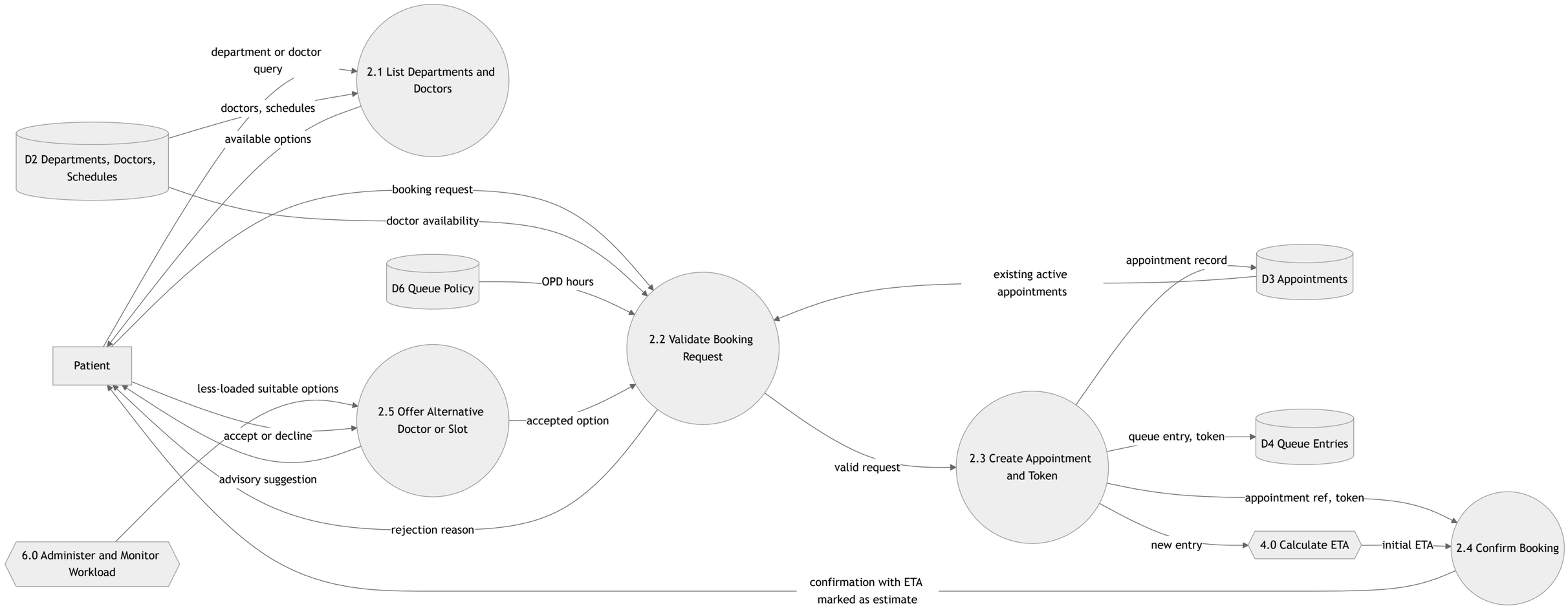

Patient
appointment ref, token, initial ETA
department, doctor, date
2.0 Book Appointment and Issue Token
appointment
D3 Appointments
ETA, last updated
new queue entry
ETA request
D4 Queue Entries
Developer / Tester
metrics report
scenario, seed, baseline
queue order
less-loaded options
7.0 Simulate and Evaluate
4.0 Calculate ETA
run config, metrics
ETA snapshot
queue length
check-in
OPD hours
D10 Simulation Runs
D7 ETA Snapshots
session or access denied
doctor availability
simulation results
clearing time
status, position, priority
registration, credentials
6.0 Administer and Monitor Workload
material event
alert status
doctor availability
observed durations, delays
departments, doctors, schedules, policy
workload dashboard, audit history, reports
policy
Administrator
D6 Queue Policy
schedules
audit trail
admin action
grace period, reinsertion and emergency rules
3.0 Operate Live Queue
ETA update
availability change
queue event
actor, action, reason
users, role assignments
check-in, skip, reinsert, emergency
queue status
call, start, complete, skip, delay, break
queue list, current patient
D2 Departments, Doctors, Schedules
D5 Queue Events and Consultations
D9 Audit Log
Front-desk Staff
Doctor
credentials
credentials
alert threshold
1.0 Manage Accounts and Access
emergency or skip notice
account, role
D1 Users and Roles
contact, channel
5.0 Send Notifications
delivery status
alert message
attempt, outcome
Notification Provider
D8 Notification Log

DFD Level 2

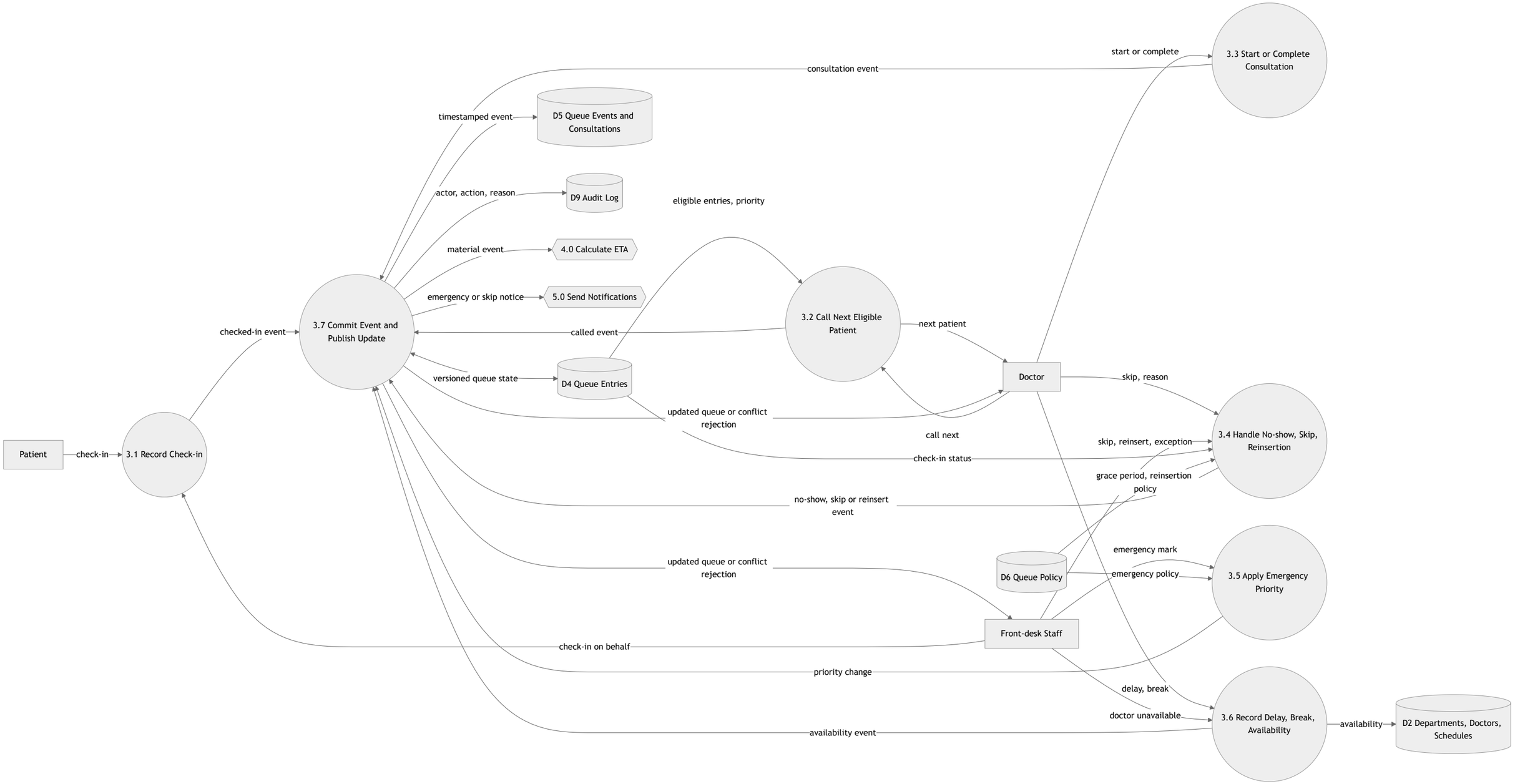
1.0 Manage Accounts and Access



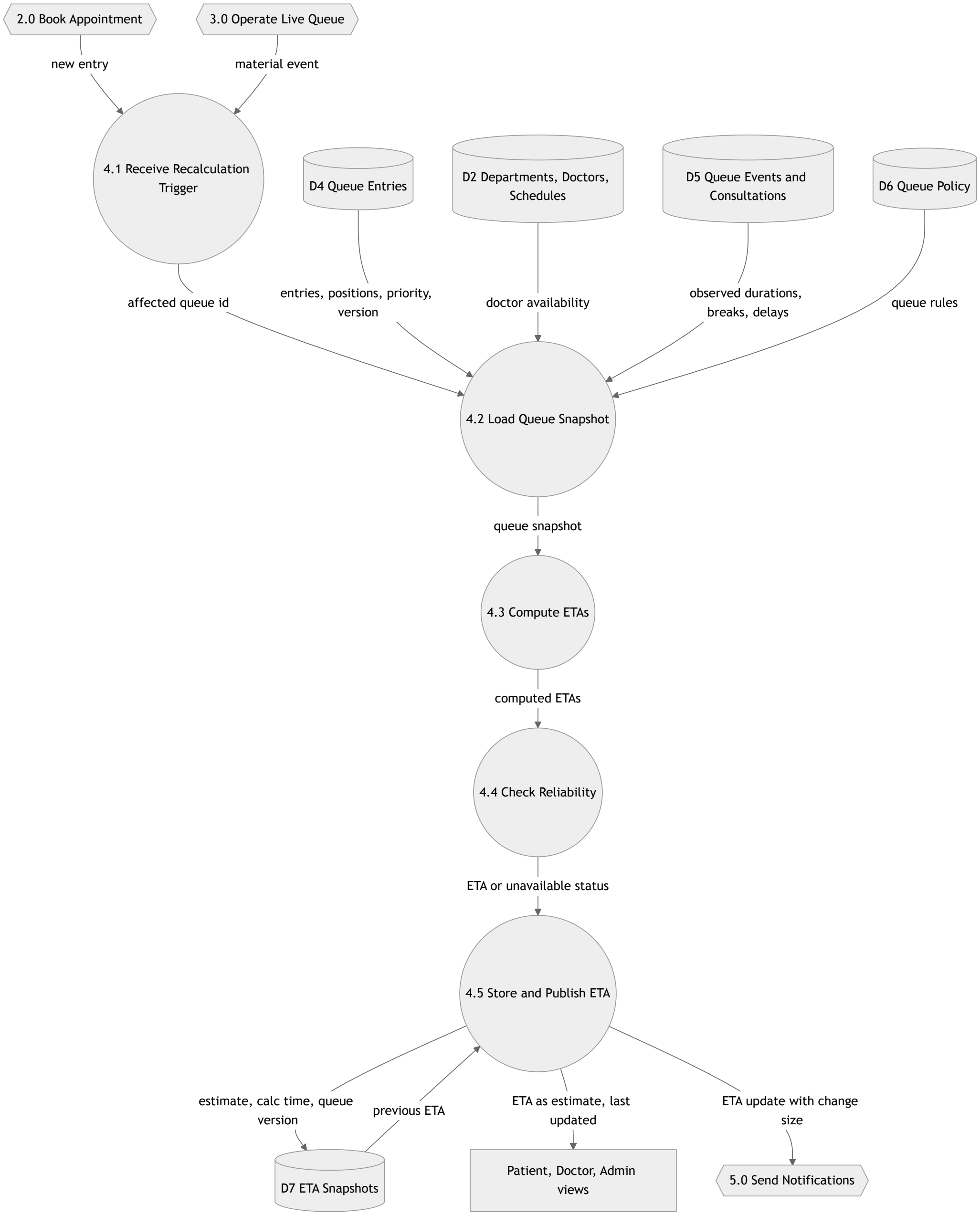
2.0 Book Appointment and Issue Token



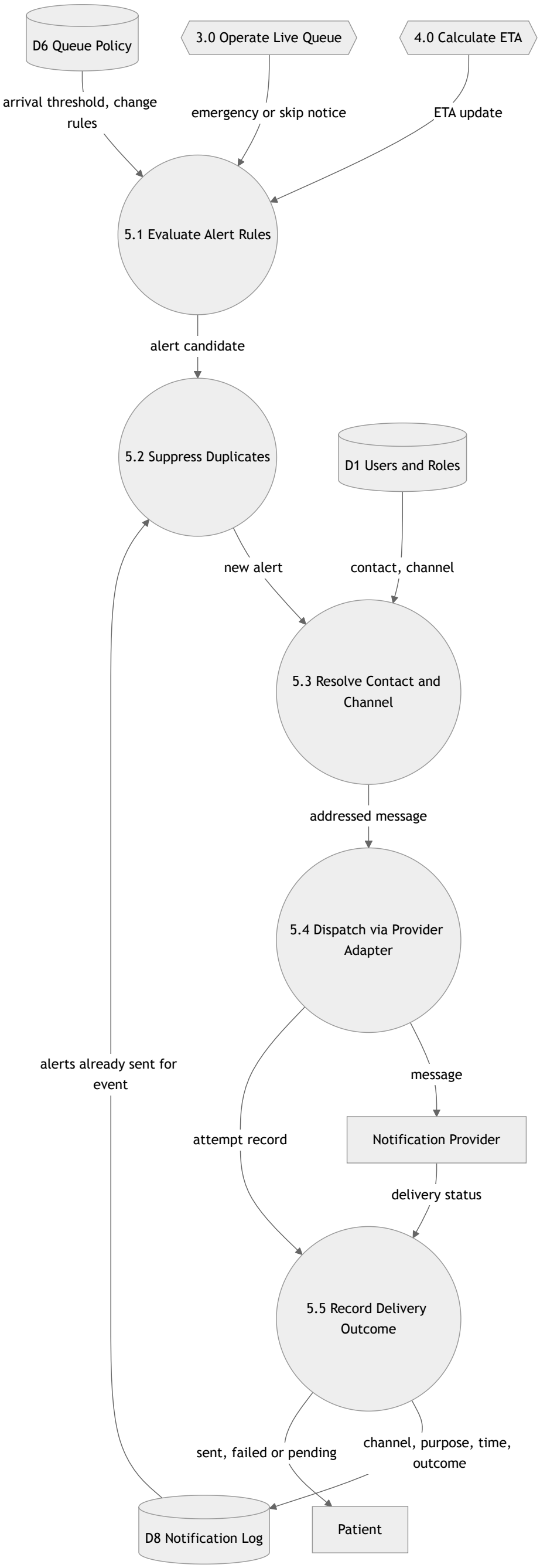
3.0 Operate Live Queue



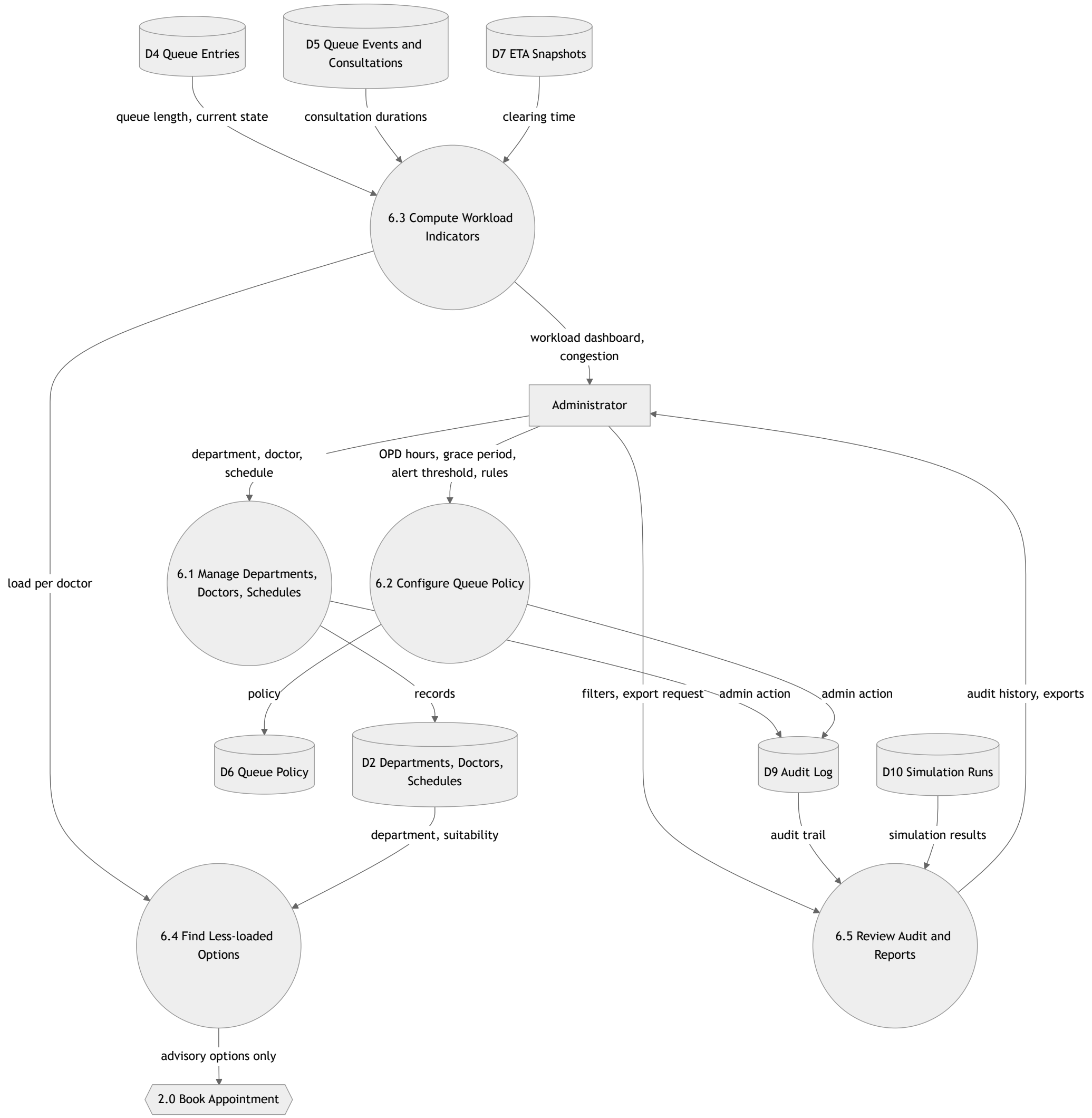
4.0 Calculate ETA



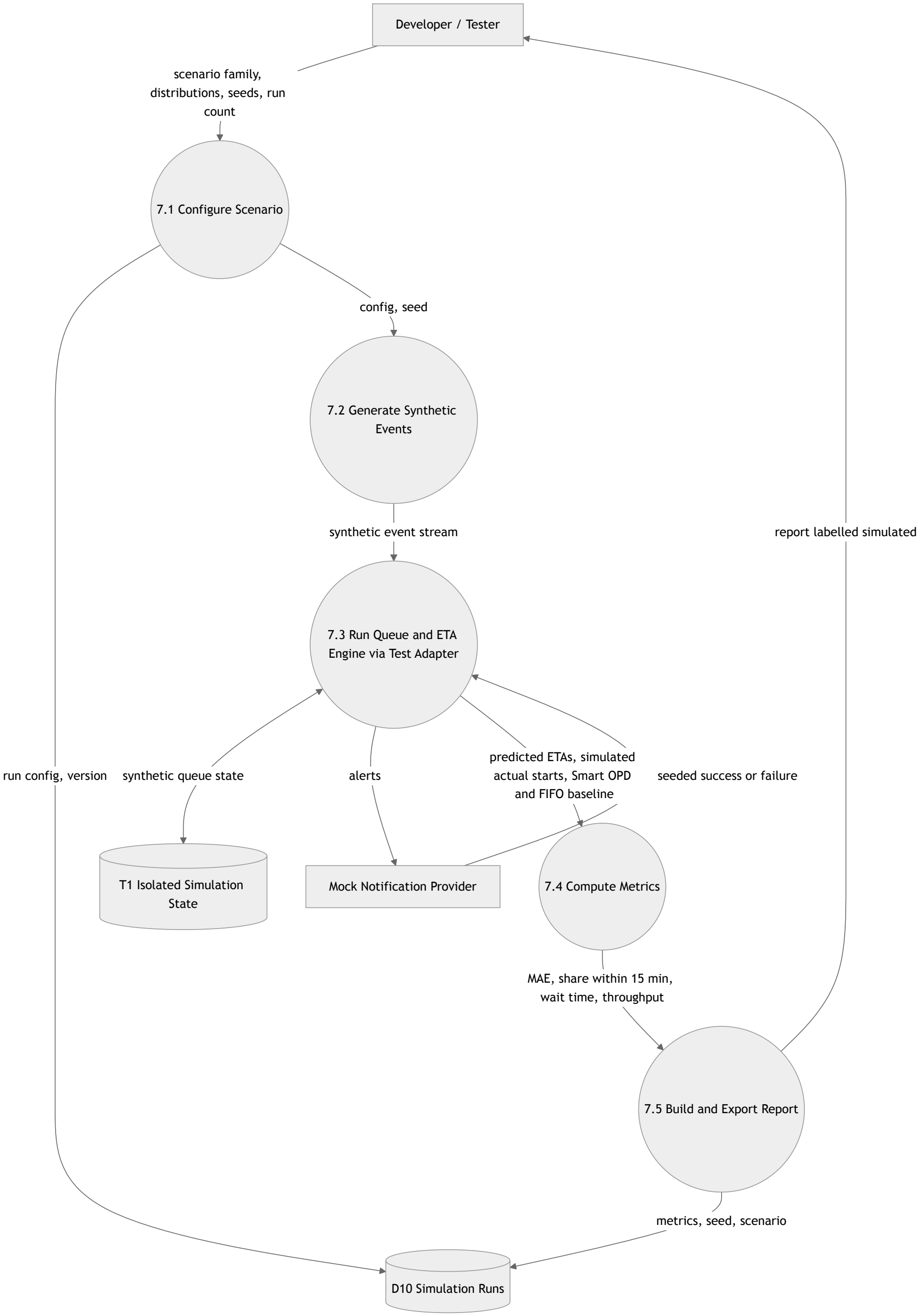
5.0 Send Notifications



6.0 Administer and Monitor Workload

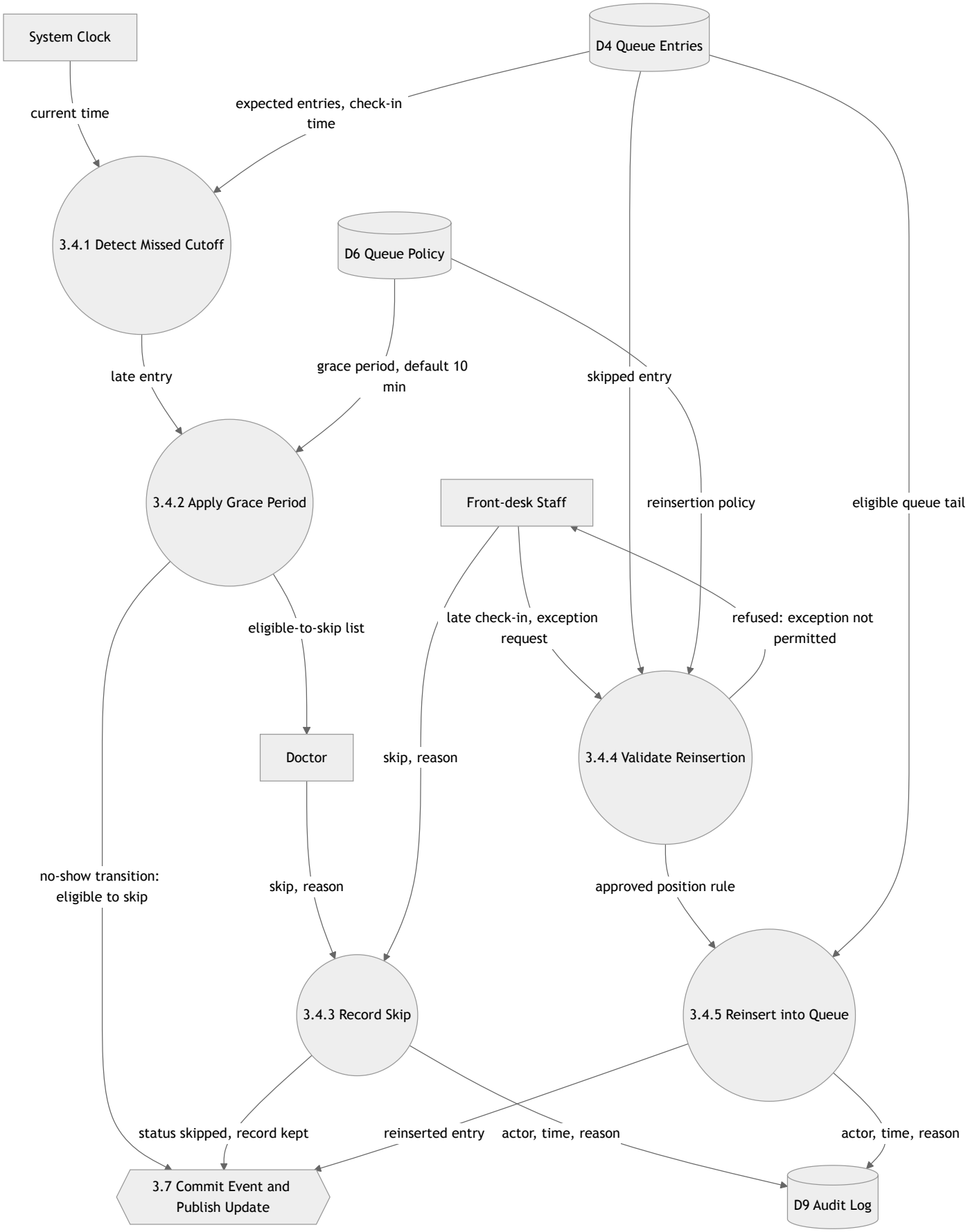


7.0 Simulate and Evaluate

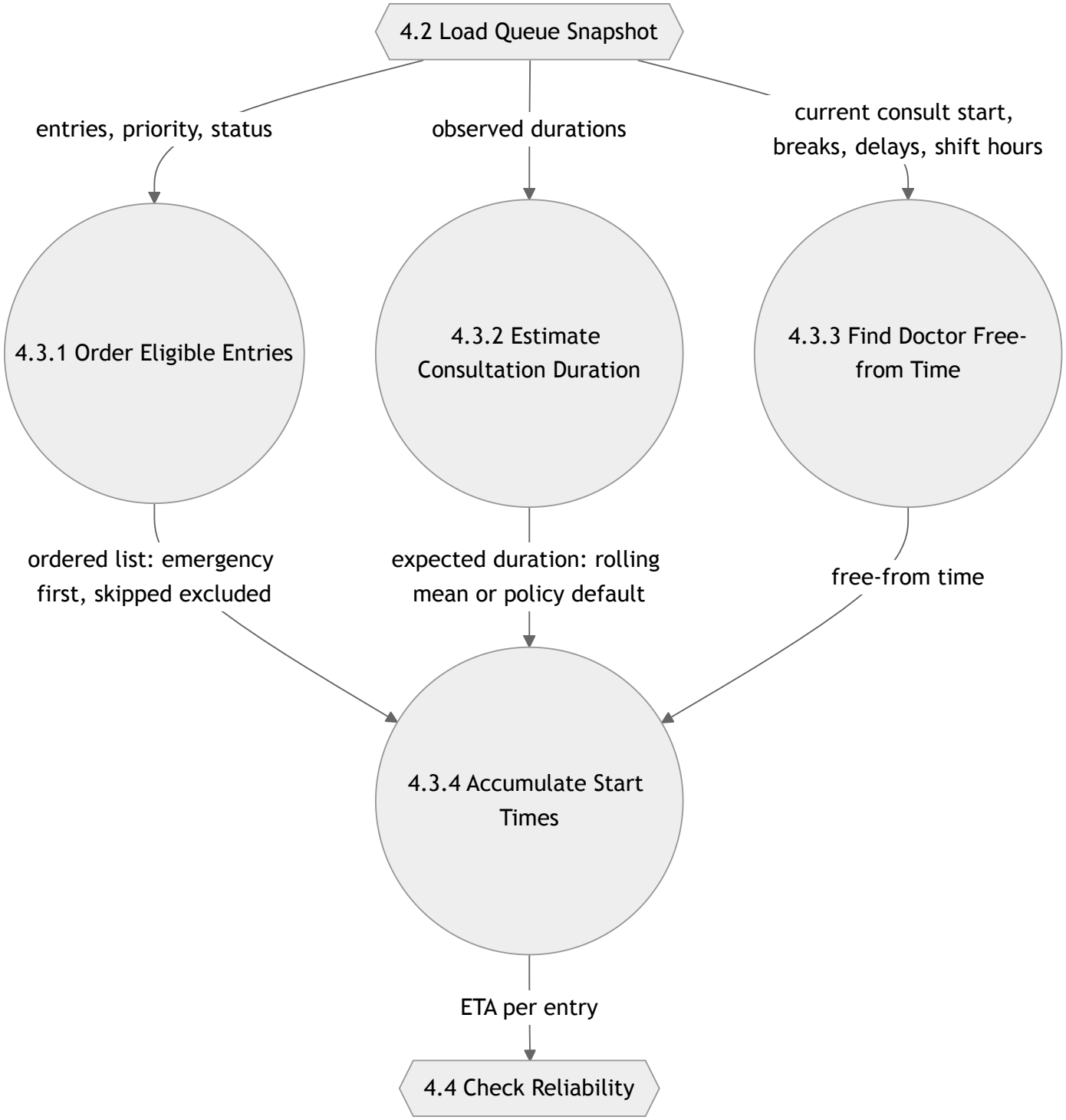


DFD Level 3

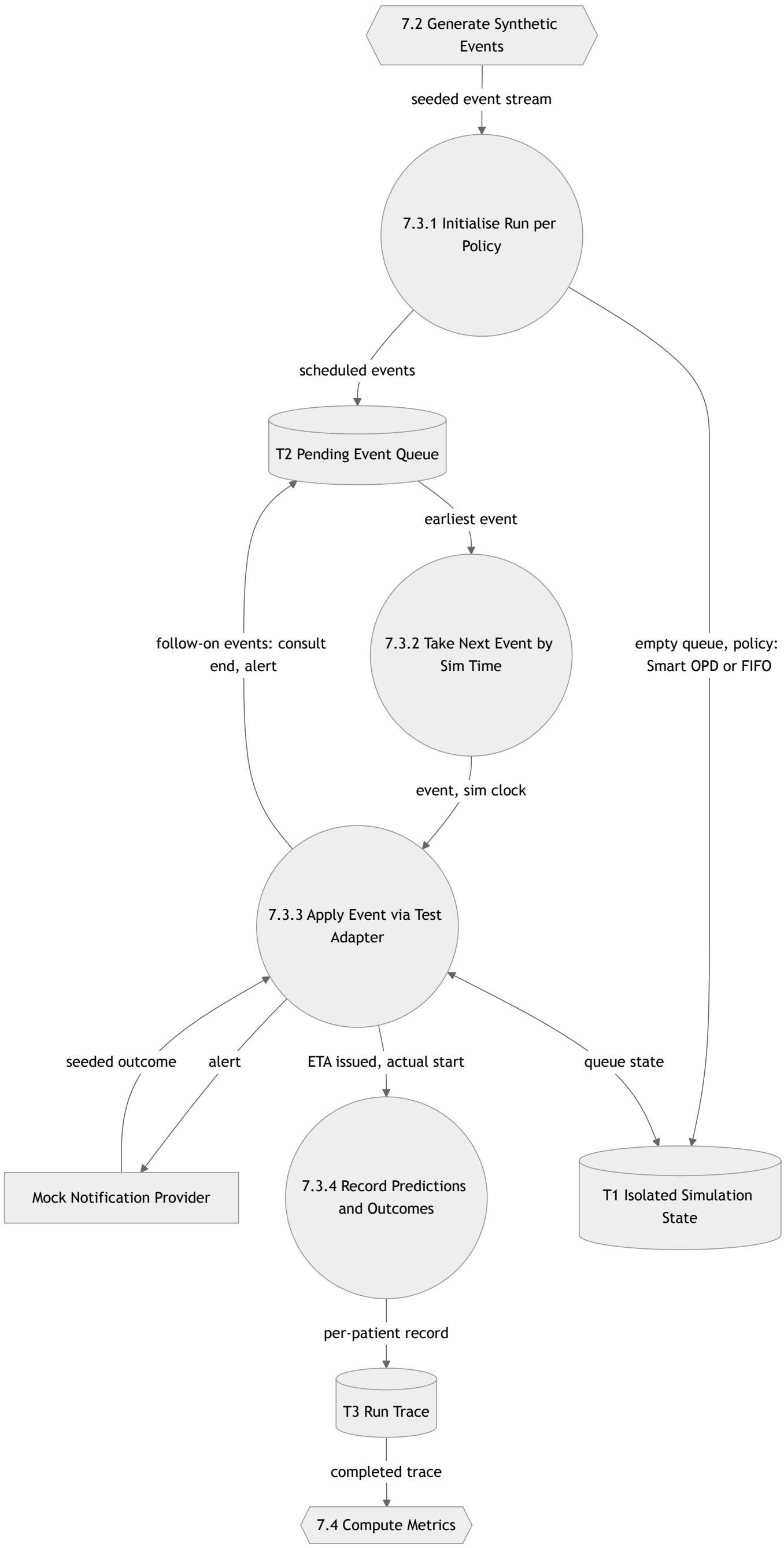
3.4 Handle No-show, Skip, Reinsertion



4.3 Compute ETAs

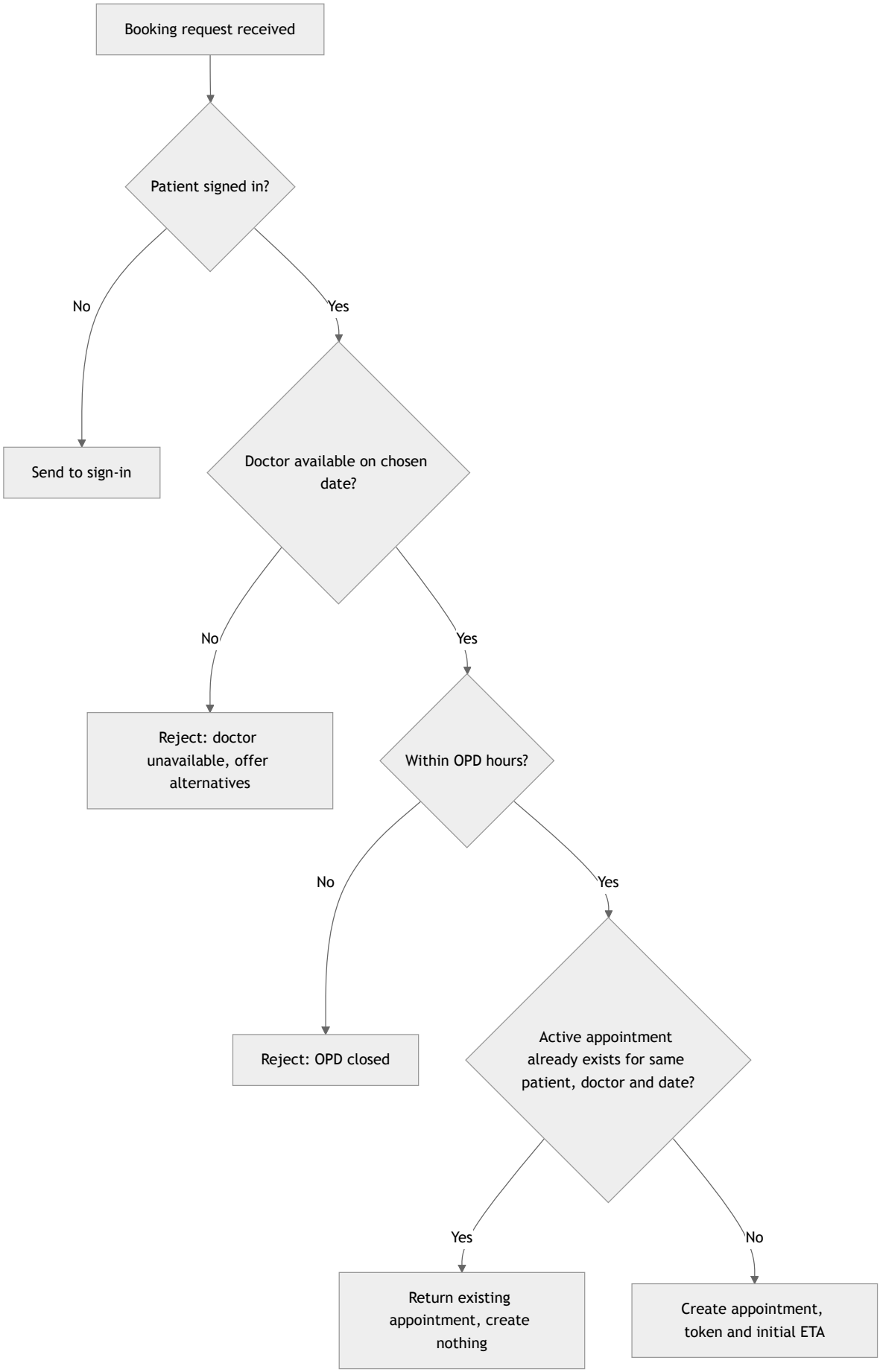


7.3 Run Queue and ETA Engine via Test Adapter

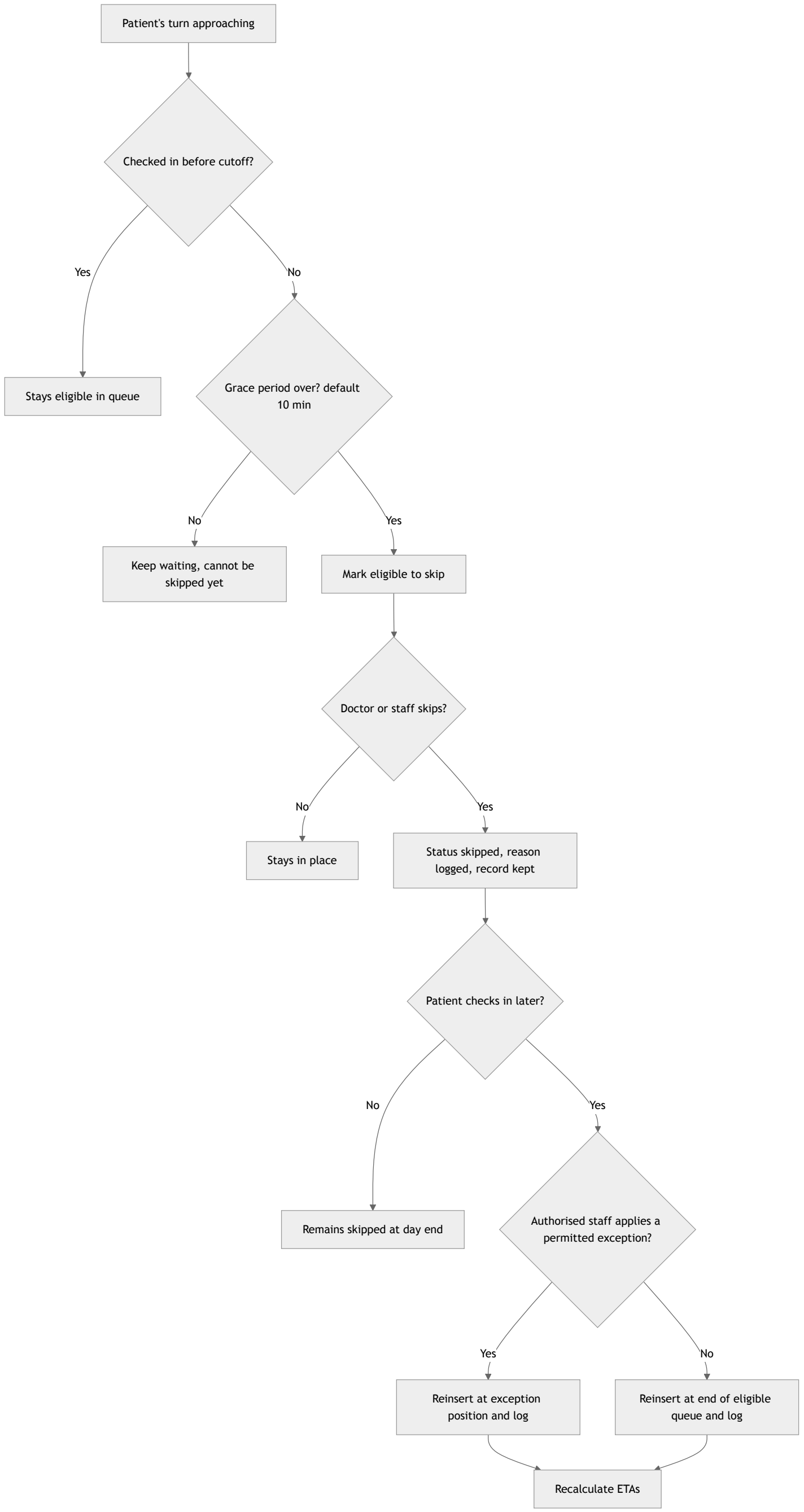


Decision Trees

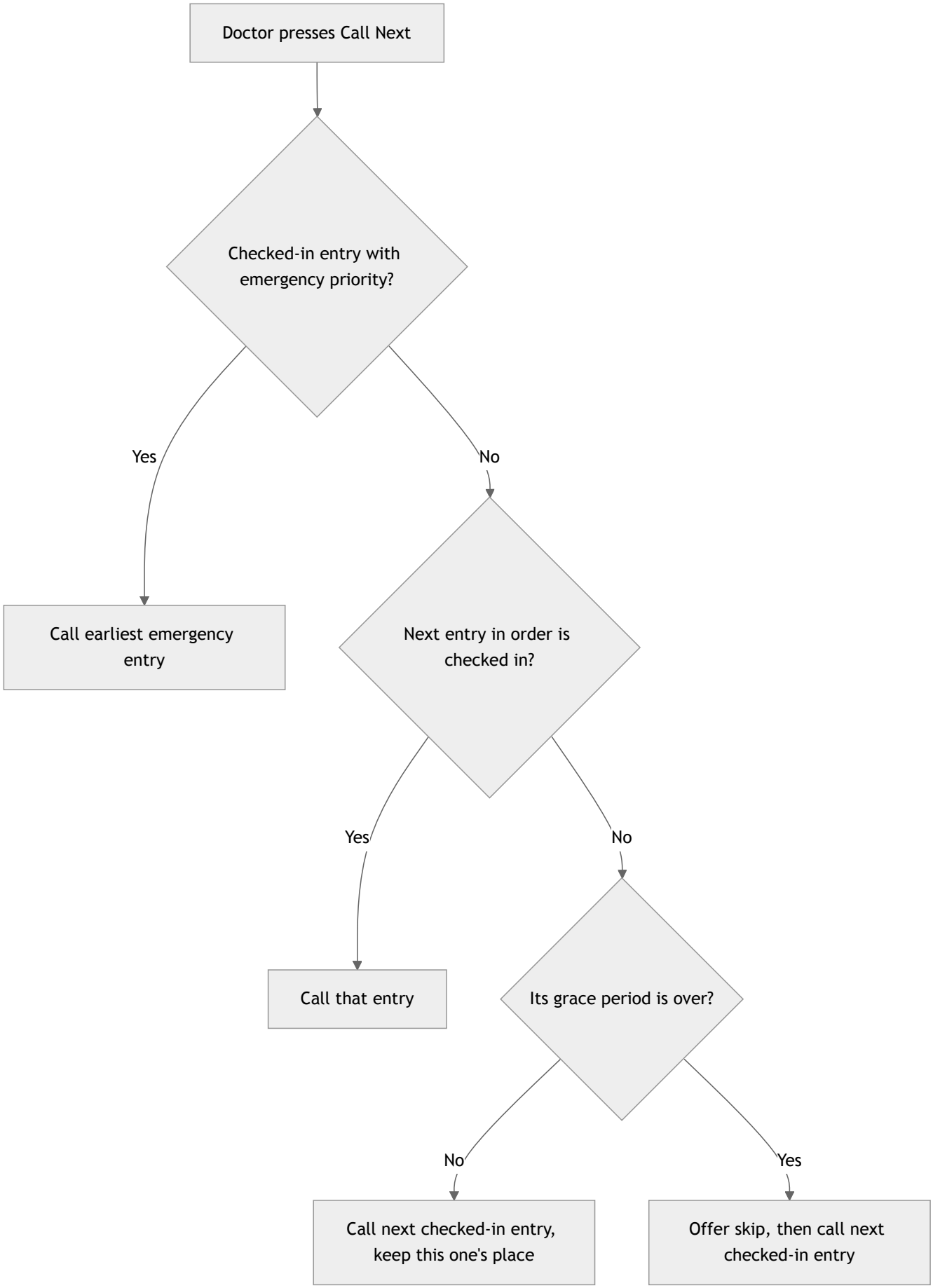
DT-1 Booking validation



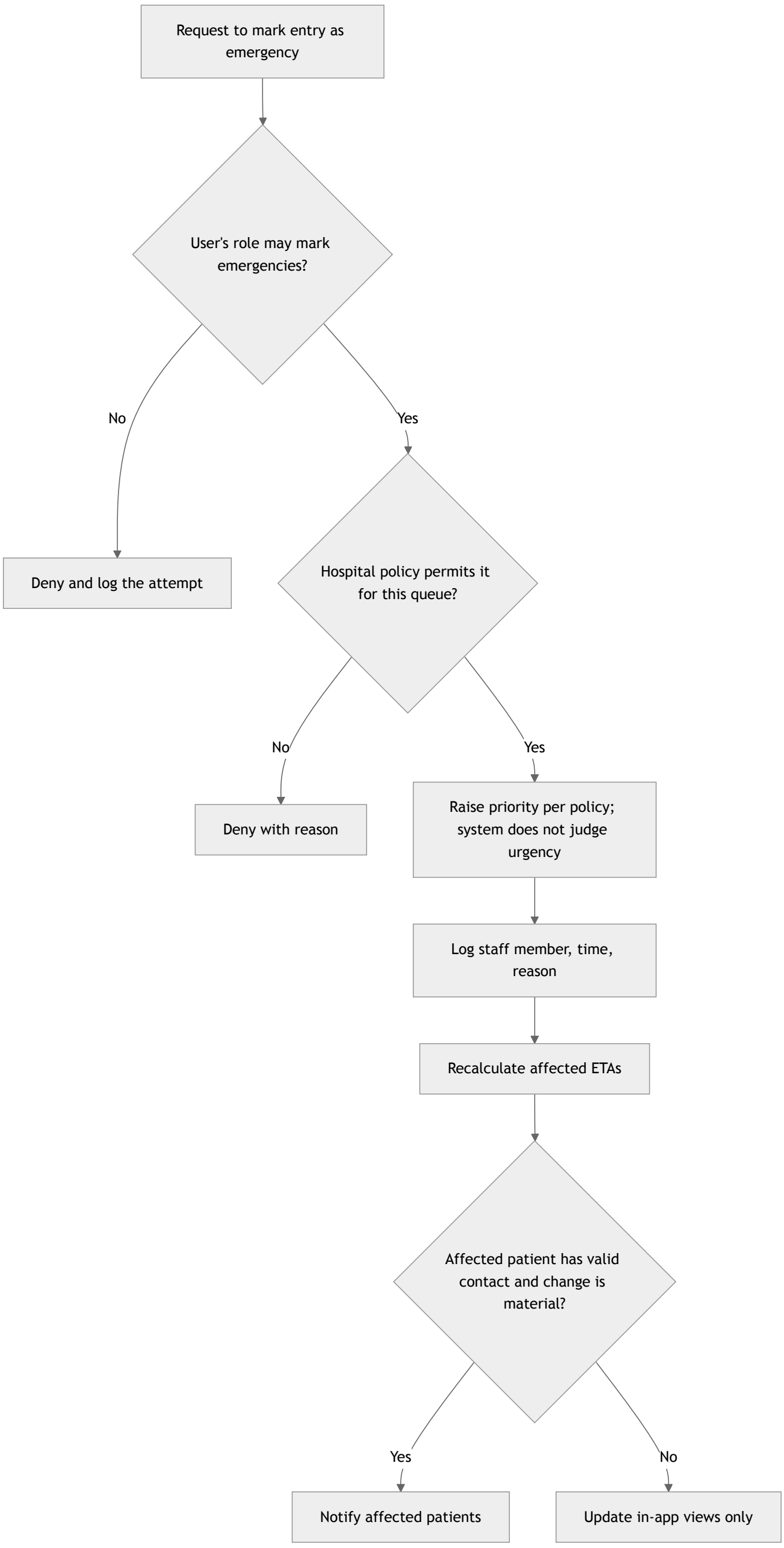
DT-2 No-show, skip and reinsertion



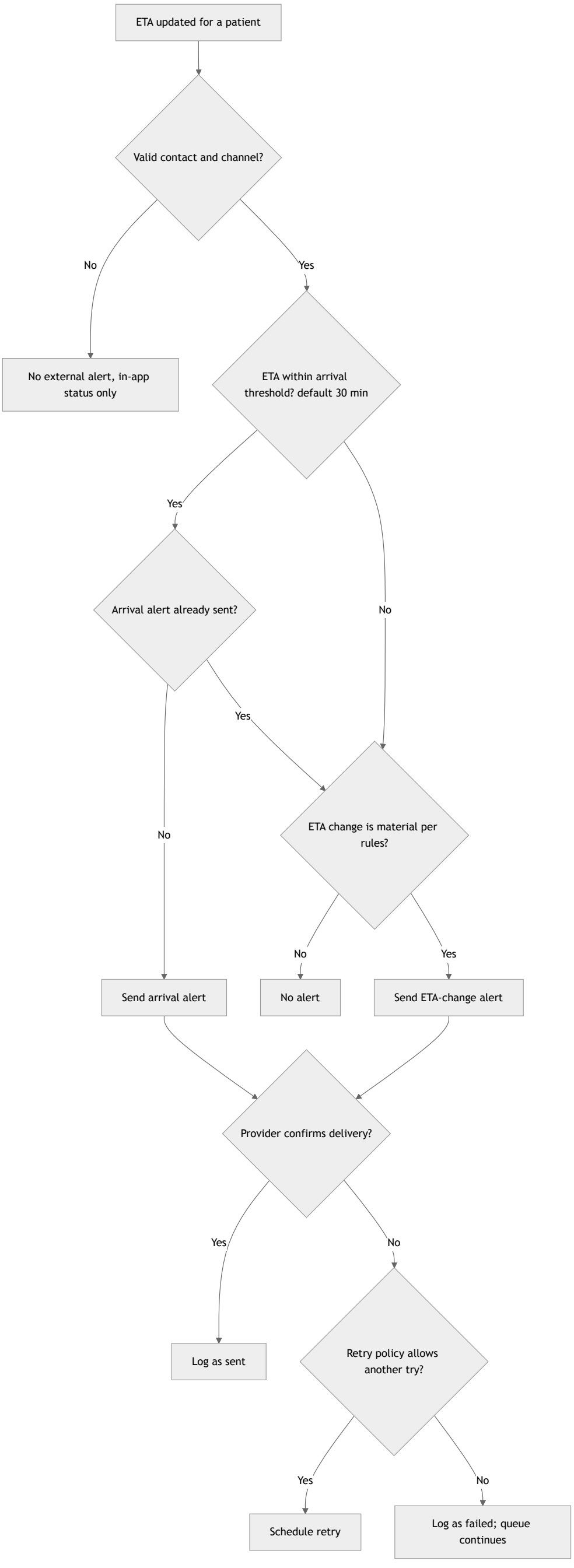
DT-3 Call next patient



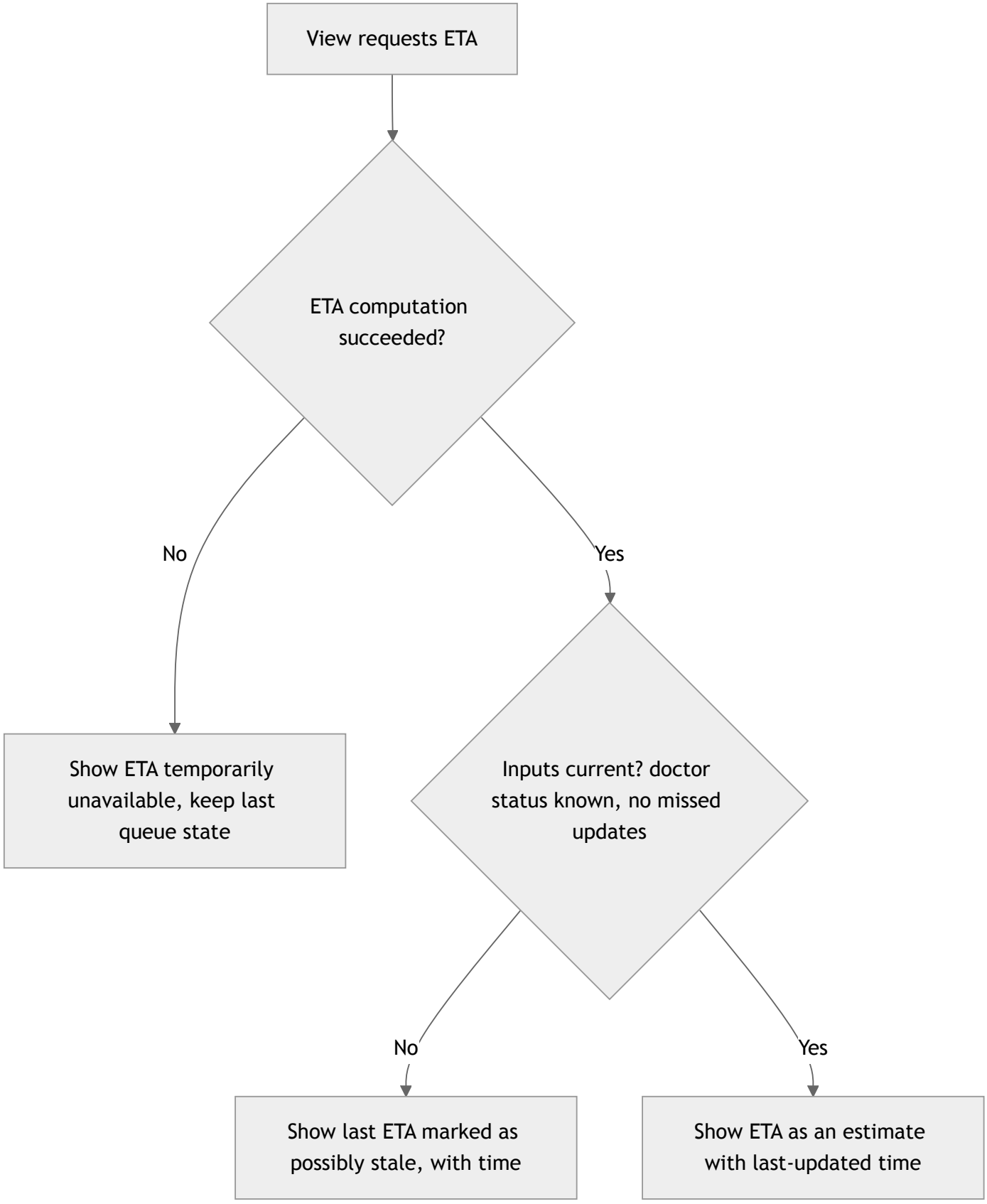
DT-4 Emergency priority



DT-5 Arrival and ETA-change alerts



DT-6 What ETA to show



Decision Tables

Table 1 · Booking validation

Conditions / Actions	R1	R2	R3	R4	R5
Conditions					
Patient signed in	N	Y	Y	Y	Y
Doctor available on date	–	N	Y	Y	Y
Within OPD hours	–	–	N	Y	Y
Duplicate active appointment exists	–	–	–	Y	N
Actions					
Send to sign-in	X				
Reject with reason		X	X		
Offer alternative doctor or slot		X			
Return existing appointment				X	
Create appointment, token, initial ETA					X

Table 2 · No-show, skip and reinsertion

Conditions / Actions	R1	R2	R3	R4	R5	R6
Conditions						
Checked in before cutoff	Y	N	N	N	N	N
Grace period over	–	N	Y	Y	Y	Y
Skipped by doctor or staff	–	–	N	Y	Y	Y
Checked in after skip	–	–	–	N	Y	Y
Authorised exception applied	–	–	–	–	N	Y
Actions						
Keep entry in its place	X	X	X			
Mark eligible to skip			X			
Set status skipped, keep record				X		
Reinsert at end of eligible queue					X	
Reinsert at exception position						X
Write audit entry (actor, time, reason)				X	X	X
Recalculate ETAs			X	X	X	X

Table 3 · Emergency priority

Conditions / Actions	R1	R2	R3	R4
Conditions				
User's role may mark emergencies	N	Y	Y	Y
Hospital policy permits it	–	N	Y	Y
Affected patients reachable and change material	–	–	Y	N
Actions				
Deny request	X	X		
Raise priority, reorder queue			X	X
Write audit entry	X		X	X
Recalculate affected ETAs			X	X
Notify affected patients			X	

Table 4 · Arrival and ETA-change alerts

Conditions / Actions	R1	R2	R3	R4	R5	R6
Conditions						
Valid contact and channel	N	Y	Y	Y	Y	Y
ETA within arrival threshold	–	Y	Y	Y	N	N
Arrival alert already sent	–	N	Y	Y	–	–
ETA change is material	–	–	N	Y	Y	N
Actions						
Show status in app only	X		X			X
Send arrival alert		X				
Send ETA-change alert				X	X	
Log attempt and delivery outcome (D8)		X		X	X	

Table 5 · Queue events and their effects

Event	Raised by	Recalc ETA	Audit entry	Order changes
Booking created	Patient (2.3)	X		X
Check-in recorded	Patient or staff (3.1)			
Consultation started	Doctor (3.3)	X		
Consultation completed	Doctor (3.3)	X		X
Delay or break recorded	Doctor (3.6)	X		
Doctor availability changed	Staff or admin (3.6, 6.1)	X	X	
No-show transition	System clock (3.4.2)	X		
Patient skipped	Doctor or staff (3.4.3)	X	X	X
Patient reinserted	Staff (3.4.5)	X	X	X
Emergency insertion	Authorised staff (3.5)	X	X	X
Queue policy changed	Admin (6.2)		X	